

Provenance-Preserving Diachronic Semantic Search via Event-Space Representations of Historical Language

Doctoral Research Proposal - Lee Goddard, June 2026

Research Aim

The project develops and evaluates an event-level approach to semantic representation in historical texts. Rather than aggregating token occurrences into type-level centroids or sense clusters, each token occurrence is treated as a discrete, provenance-bearing contextual event in embedding space. The project employs the MacBERTh Early Modern English model initially on the EEBO-TCP corpus from 1625–1689, later expanding to other corpora, to test whether this preserves visibility of semantic instability, polysemy, and contestation that aggregation methods suppress.

The ultimate deliverable is a working diachronic semantic search interface: users query concepts in modern English and retrieve historically appropriate source passages, with full provenance links enabling seamless transition from distant to close reading.

Central hypothesis: Neighbourhood geometry in the full event space, when analysed without premature aggregation, better reveals the dynamics of historical meaning-making, particularly in periods of rapid rhetorical and conceptual change.

Background and Motivation

Keyword search in historical corpora systematically fails through false negatives (unanticipated vocabulary) and false positives (polysemy across time), as documented by McGillivray, Nanni and Beelen (2023). While semantic change detection has advanced (SemEval-2020, DWUG, Tahmasebi et al. 2021), no system support actual meaning-centred, time-aware search linked back to source passages.

Existing methods typically collapse occurrences into aggregated representations, losing provenance and allowing dominant contextual attractors to obscure rarer or contested usages. The polemical pamphlet literature of the Civil War, Interregnum, Restoration, and Glorious Revolution offers an ideal test case of documented semantic instability around terms such as *liberty*, *conscience*, *authority*, and *sovereignty*.

Methodology: a tiered, provenance-preserving pipeline

Tier 1 – Contextual Event Construction Parse EEBO-TCP XML into a relational database with full metadata and positional information. Generate contextual embeddings for every content token using MacBERTh. Store events in a Zarr store with dual identities (lexical occurrence + contextual observation) and complete provenance links, allowing multiple geometric positions for the same textual locus under different context windows.

Tier 2 – Event-Space Neighbourhood Analysis Build a FAISS index over the full event population. Analyse neighbourhoods directly (overlap, entropy, token/document concentration, temporal distribution) without aggregating into clusters or centroids. This extends the DWUG usage-graph approach from sampled, annotated pairs to the complete corpus while preserving provenance.

Tier 3 – Diachronic Bridge and Search Interface Align MacBERTh embeddings with a modern English transformer (eg BERT) via Procrustes transformation, using texts from their temporal overlap and stable anchor vocabulary. The resulting bridge supports two interface modes:

- **Modern-concept query:** Project modern query into aligned "bridge" space; retrieve and visualise historical sense neighbourhoods; user selects from clusters/text samples; ranked source passages with characteristic period vocabulary.
- **Period-vocabulary query:** Direct MacBERTh query alignment and neighbourhood retrieval, tracking evolution over time.

All results maintain direct links to source passages. An optional extension explores multi-scale context modelling (micro-window vs clause vs paragraph vs broader discourse windows) to distinguish local rhetorical usage from global discursive repositioning.

Advancing the State of the Art

Computational Linguistics/NLP: Introduces event-level (rather than aggregated) analysis for semantic change and search; novel use of MacBERTh's extended training range for empirically grounded cross-period alignment; multi-scale context modelling.

Digital Humanities: Delivers the first practical prototype of diachronic semantic search over a major historical corpus, directly addressing McGillivray et al's call, with full provenance.

Early Modern / Intellectual History: Provides new tools for studying conceptual contestation in 17th-century political discourse, bridging computational neighbourhood analysis with the close-reading traditions of Skinner, Pocock, and Koselleck.

Expected Outputs

- Doctoral thesis with methodological contribution and empirical findings on semantic contestation in EEBO pamphlets.
- Reusable provenance-preserving event store and FAISS index.
- Open-source, researcher-facing diachronic semantic search interface.
- Peer-reviewed publications (eg Computational Approaches to Historical Language Change workshop, DH venues).
- Open-source codebase for event extraction, diagnostics, alignment, and search.
- Documentation and examples of running the codebase on further corpora.

Current Progress

A functional prototype pipeline already exists: TEI XML - RDBMS - MacBERTh event extraction - Zarr provenance store - FAISS indexing - neighbourhood diagnostics and visualisation. Preliminary results confirm dominant attractor structures, and demonstrate that event-level analysis can recover distinct contextual clusters (eg *church* as building, institution, or office) that aggregation tends to obscure.